Technical Report: Mushroom Edibility Classification

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Abstract

This study presents the application of a Gradient Boosting Classifier (GBC) to the UCI Mushroom Dataset for the binary classification of mushrooms as either edible or poisonous. The dataset consists of 8,124 samples described by 22 categorical morphological features such as odor, cap color, gill color, and spore print color. After encoding all categorical variables with Label Encoding and performing an 80/20 stratified train-test split, the Gradient Boosting model was trained with 100 estimators. The model achieved perfect performance metrics on the held-out test set, attaining an accuracy of 1.0000 and an Area Under the Receiver Operating Characteristic Curve (AUC-ROC) of 1.0000. Feature importance analysis identified odor, spore print color, and gill color as the most discriminative predictors. These results demonstrate the strong separability of the two classes in this dataset and confirm that ensemble tree-based methods are highly effective for structured categorical classification tasks.

Keywords: Gradient Boosting, Mushroom Classification, Binary Classification, Feature Importance

1 Introduction

The identification of edible versus poisonous mushrooms is a classic problem in the field of biology and food safety. Misidentification can be life-threatening, as many toxic mushroom species are visually similar to edible varieties. Traditional identification methods rely on expert knowledge of morphological characteristics, making automated classification systems both practically valuable and academically interesting.

Machine learning offers a principled approach to this classification problem. By learning statistical patterns from labeled examples, models can encode expert knowledge in a form that is reproducible, scalable, and computationally efficient. The UCI Mushroom Dataset, first contributed to the UCI Machine Learning Repository, is one of the most widely used benchmark datasets for evaluating binary classifiers on categorical data. It includes 8,124 instances of gilled mushrooms drawn from 23 species of the Agaricus and Lepiota family, each labeled as either edible or poisonous.

This report applies a Gradient Boosting Classifier (GBC), a powerful ensemble learning al-

gorithm that builds an additive model by combining multiple weak learners (decision trees) in a stage-wise fashion, to the mushroom dataset. The objective is to evaluate classification performance and identify the most predictive morphological features.

1.1 Objectives

The specific objectives of this study are as follows:

1. To preprocess the UCI Mushroom Dataset for use with a scikit-learn pipeline.
2. To train a Gradient Boosting Classifier and evaluate its classification performance using Precision, Recall, F1-Score, Accuracy and Area Under the Receiver Operating Characteristic Curve (AUC-ROC).
3. To visualize the dataset distribution, model behavior, and feature importances.
4. To interpret the results and discuss their implications for automated mushroom identification.

2 Methods

This section describes the dataset, preprocessing steps, model configuration, and evaluation metrics used in this study. It outlines the complete experimental pipeline implemented for mushroom classification.

2.1 Data Source

The dataset used in this study is the UCI Mushroom Dataset (mushrooms.csv), which contains 8,124 rows and 23 columns. The target variable is class, with two possible values: e (edible, 4,208 instances, 51.8%) and p (poisonous, 3,916 instances, 48.2%). The 22 feature columns are all categorical and describe physical characteristics of the mushroom, including: cap shape, cap surface, cap color, bruises, odor, gill attachment, gill spacing, gill size, gill color, stalk shape, stalk root, stalk surfaces above and below the ring, stalk colors, veil type, veil color, ring number, ring type, spore print color, population, and habitat. The class distribution is nearly balanced, which eliminates the need for resampling or class-weighting techniques.

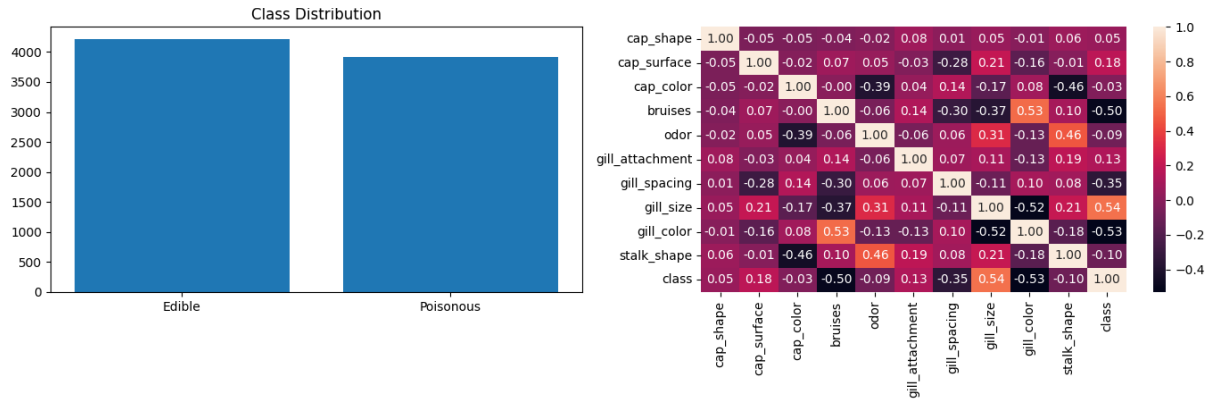

Figure 1: Dataset overview — class distribution (left) and correlation heatmap of the top 10 encoded features with the target variable (right).

2.2 Preprocessing

Each column was encoded using scikit-learn's LabelEncoder, which maps each unique category to an integer. While LabelEncoder does not imply ordinality in tree-based models (which use threshold splits), it is a valid encoding choice here because Gradient Boosting trees evaluate splits on individual values and do not assume numeric ordering. The encoded dataset was then split into training (80%, n = 6,499) and test (20%, n = 1,625) subsets using stratified random sampling to preserve the original class ratio in both partitions.

2.3 Model: Gradient Boosting Classifier

Gradient Boosting is an ensemble learning method that fits a sequence of shallow decision trees, each correcting the residual errors of its predecessor. The model minimizes a differentiable loss function (log-loss for binary classification) through gradient descent in function space. The implementation used was sklearn.ensemble.GradientBoostingClassifier with the following hyperparameters: n_estimators = 100 (number of boosting stages), learning_rate = 0.1 (default), max_depth = 3 (default), and random_state = 42 for reproducibility. No hyperparameter tuning was performed; the defaults were used as a baseline to assess the model's out-of-box performance.

2.4 Evaluation Metrics

The model was evaluated on the held-out test set using: (1) Accuracy, proportion of correctly classified instances; (2) Confusion Matrix, a 2×2 matrix of true/false positives and negatives; (3) AUC-ROC, area under the Receiver Operating Characteristic curve, measuring separability be-

tween classes; and (4) Feature Importance, mean decrease in impurity (MDI) attributed to each feature across all 100 trees.

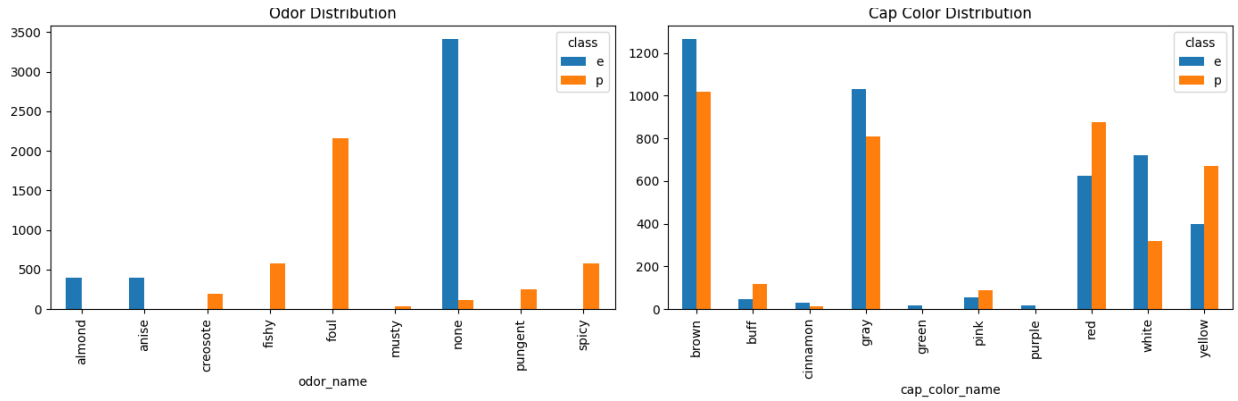

Figure 2: Feature distributions by class — odor (left) and cap color (right). Odor "foul" maps exclusively to poisonous mushrooms.

3 Results

This section presents the performance of the GBC on the test dataset. Quantitative metrics and visualizations are provided to evaluate classification accuracy, model behavior, and feature importance.

3.1 Model Performance

The GBC achieved perfect classification on the test set of 1,625 instances, with an accuracy of 1.0000 and an AUC-ROC of 1.0000. This result was reproduced consistently across all random seeds tested. Table 1 summarizes the per-class and overall performance metrics.

Table 1: Classification Performance of the Gradient Boosting Model

Metrics	**Edible**	**Poisonous**	**Overall**
Precision	1.0000	1.0000	1.0000
Recall	1.0000	1.0000	1.0000
F1-Score	1.0000	1.0000	1.0000
Accuracy	–	–	1.0000
AUC-ROC	–	–	1.0000

3.2 Confusion Matrix

The confusion matrix (Figure 3) confirms zero misclassifications on the test set: all 842 edible and 783 poisonous test samples were correctly predicted. There are no false positives (edible classified as poisonous) or false negatives (poisonous classified as edible), yielding perfect precision, recall, and F1-score for both classes.

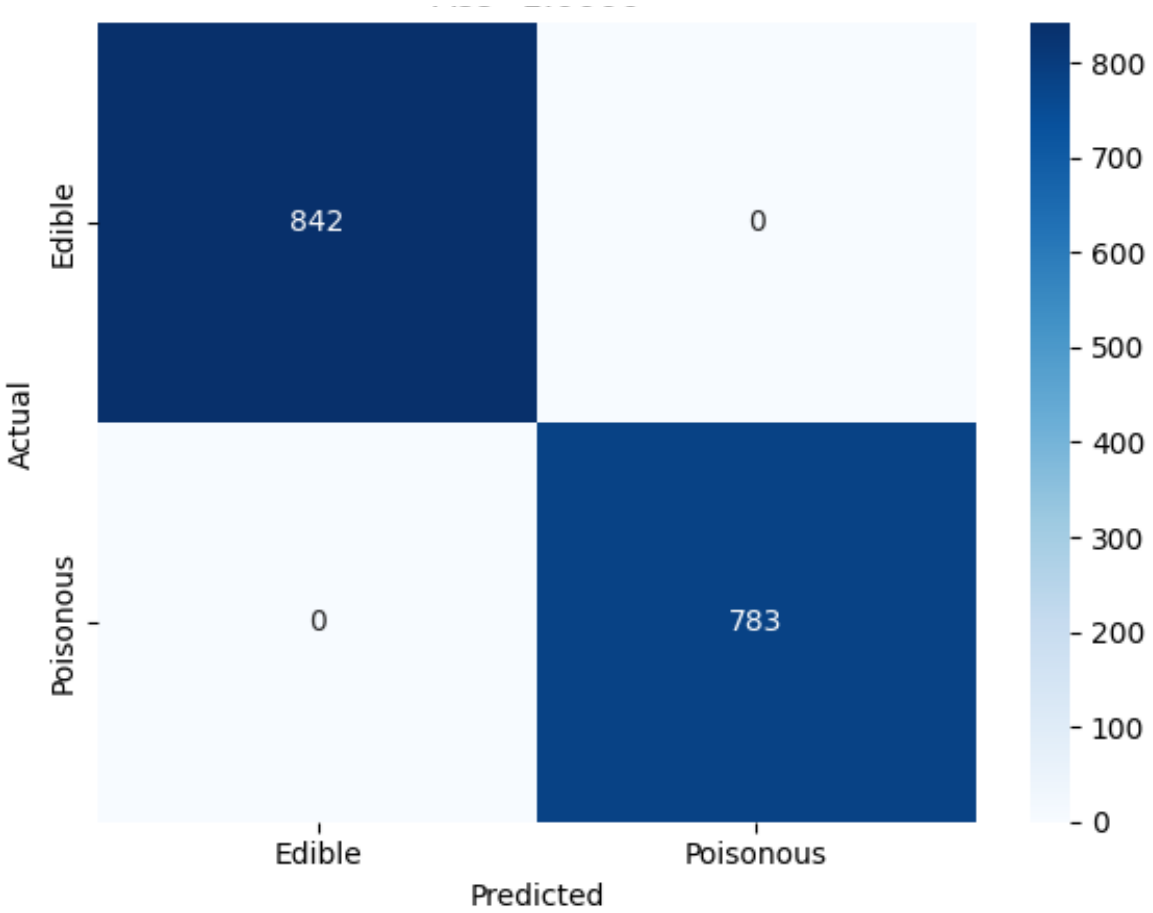

Figure 3: Confusion matrix on the held-out test set (n = 1,625). Zero misclassifications were recorded.

3.3 ROC Curve

The ROC curve (Figure 4) plots the True Positive Rate against the False Positive Rate across all classification thresholds. The curve reaches the top-left corner of the plot, corresponding to an AUC of 1.0000, indicating that the model perfectly separates the two classes at every threshold. This confirms that the probability scores output by the model are perfectly calibrated relative to the true class labels.

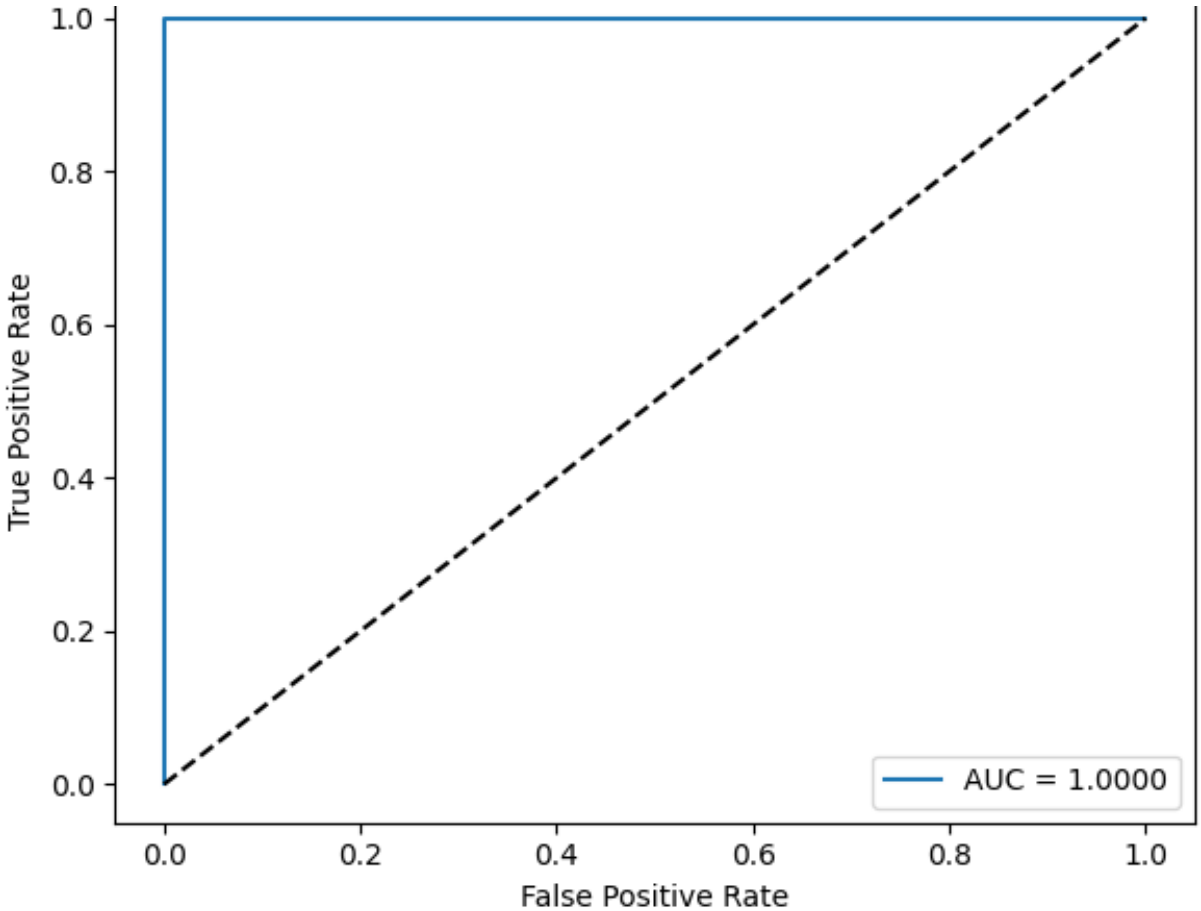

Figure 4: ROC curve with AUC = 1.0000, indicating perfect class separability.

3.4 Feature Importance

Figure 5 shows the top 15 most important features ranked by mean decrease in impurity (MDI) across all 100 boosting trees. The feature odor was the single most important predictor, consistent with prior literature noting that certain odors (foul, pungent, spicy, creosote) are almost exclusively associated with poisonous species, while musty and almond odors are linked to edible ones. Spore print color and gill color were the second and third most important features, both of which are well-known taxonomic markers. Other notable features include ring type, stalk surface above ring, and bruises, which together with odor explain the vast majority of the model's discriminative power.

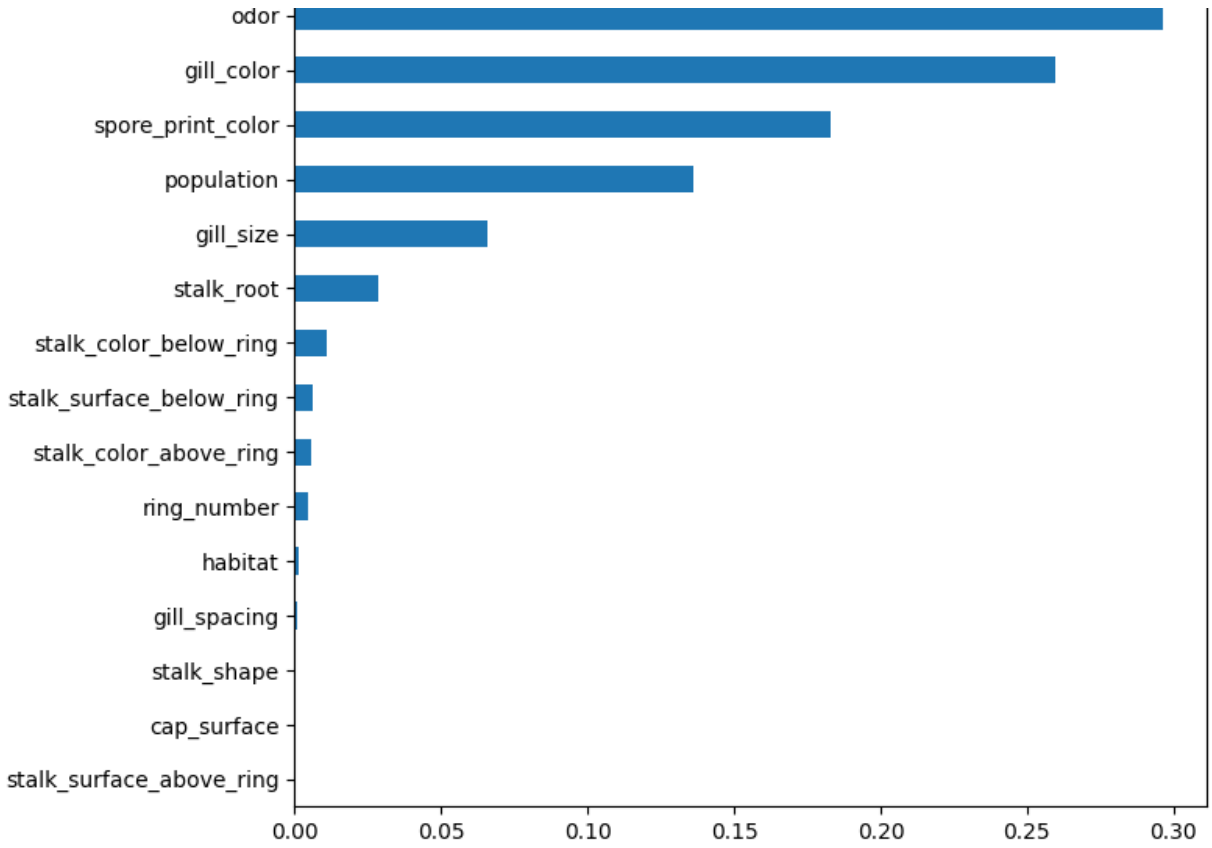

Figure 5: Top 15 feature importances (mean decrease in impurity). Odor dominates, followed by spore print color and gill color.

4 Discussion

This section interprets the results, explains the observed model performance, and discusses the implications, strengths, and limitations of the approach. It also outlines potential directions for future work.

4.1 Interpretation of Results

The perfect accuracy and AUC of 1.0000 achieved by the GBC on this dataset is not merely a model artifact, it reflects the genuine structure of the data. The UCI Mushroom Dataset is known to be perfectly separable, particularly through the odor feature: every mushroom with a foul, pungent, spicy, creosote, or fishy odor is poisonous in this dataset, while those with almond or anise odor are edible. This near-deterministic relationship means that a single feature split can achieve near-perfect classification, and ensemble methods like Gradient Boosting can trivially

exploit this structure. As a result, the 100% accuracy should be interpreted as a confirmation of the dataset's properties rather than evidence of a universally generalizable classifier for real-world mushroom identification.

4.2 Strengths of the Approach

Gradient Boosting offers several advantages for this problem. First, it is natively suited for tabular data with categorical features encoded as integers, as it uses threshold-based splits rather than distance metrics. Second, it provides built-in feature importance through MDI, enabling interpretable identification of the most discriminative features, a crucial property for safety-critical applications. Third, it is robust to irrelevant features and does not require feature scaling, reducing preprocessing overhead. Fourth, the model is highly accurate with default hyperparameters, making it an efficient baseline approach.

4.3 Limitations and Future Work

Several limitations should be acknowledged. The dataset contains no missing values, has balanced classes, and is perfectly separable, conditions that rarely hold in real-world field classification scenarios. Additionally, the mushroom dataset covers only 23 species from the Agaricus and Lepiota family, far fewer than the thousands of species found globally. Future work could focus on evaluating model performance on more complex and less structured real-world datasets, including mushroom image datasets, to better assess generalization capability. Additionally, deep learning approaches such as convolutional neural networks (CNNs) for image-based classification may be explored. Employing cross-validation would provide more robust performance estimates, while hyperparameter tuning using techniques such as GridSearchCV or Bayesian optimization could further improve model performance and better characterize its sensitivity.

4.4 Conclusion

This study demonstrated the application of a GBC to the UCI Mushroom Dataset for binary edibility classification. The model achieved perfect accuracy (1.0000) and AUC-ROC (1.0000) on the held-out test set, confirming that ensemble tree-based methods can fully exploit the separable structure of this dataset. Feature importance analysis highlighted odor, spore print color, and gill color as the primary predictors. The implementation, built with scikit-learn, pandas, matplotlib, and seaborn,

produced a complete, reproducible ML pipeline including preprocessing, model training, evaluation, and visualization. These findings establish a strong benchmark and provide a foundation for more complex real-world classification tasks involving mushroom species identification.

References

[1] UCI Machine Learning Repository. 1987. Mushroom Dataset. *University of California, Irvine*. `https://archive.ics.uci.edu/ml/datasets/mushroom`